

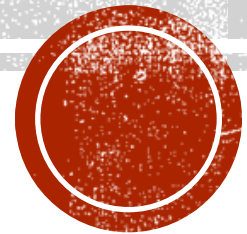
UNAUTHORIZED SYSTEM ACCESS AND MODIFICATION

Case ID: Group 3

Investigators: Rethick, Sreenivas, Rahgu, Bhargav

Date: 08-11-2024

Acquisition Date: 25-10-2024



Given Statement

- ◇ One day an IT dept. guy/gal came to work to find the message below written in a file on his/her system! He/Her immediately reported that to you. Now you must uncover how it happened
- ◇ Obtained artifacts
 - ◇ System Image File
 - ◇ Hashes and password text file

Hello master,

Catch me if you can!

Best regards,
Your Admin ;)



OBJECTIVE

- ◇ To investigate unauthorized access to a Windows 8.1 Enterprise Edition system, identify evidence of an attack, and analyze artifacts to determine the scope and actions of the attacker.

EVIDENCE SUMMARY

- Image Information:
 - Acquisition started: Fri Dec 11 09:35:03 2015
 - Acquisition finished: Fri Dec 11 09:47:19 2015
- Image Verification Results:
 - Verification started: Fri Oct 25 09:47:19 2024
 - Verification finished: Fri Oct 25 09:52:48 2024
 - MD5 checksum: 9a737b6a21ecbe979126c5661ca7b8db : **verified**
 - SHA1 checksum: ebef2bdc6329563ccf85769c3e5b8ceca852992b : **verified**



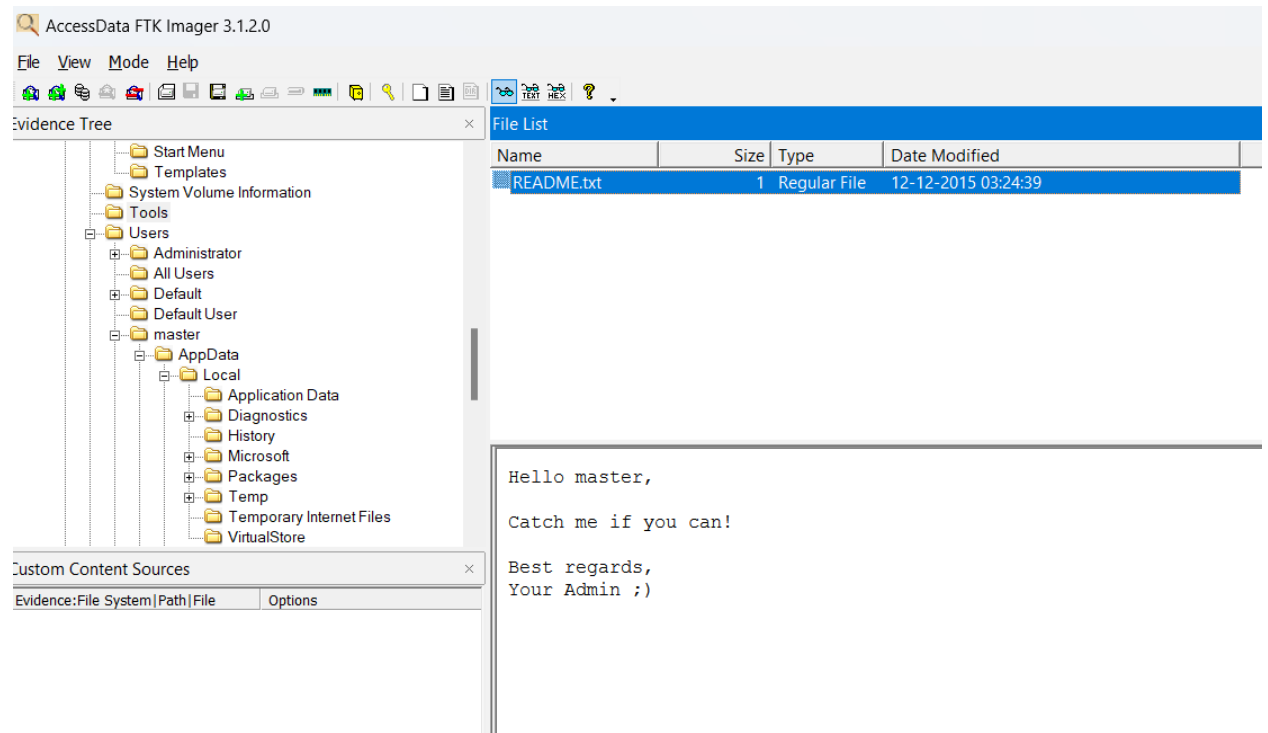
TOOLS & SOFTWARES USED

- FTK Imager: Used for mounting and analyzing image files.
- Shellbag Explorer: Used to analyze the UserClass.dat file.
- Registry Explorer: Used for examining registry files.
- Virus Total: Used for analyzing potentially malicious files.
- VM: Sandbox Environment is used to safely examine suspicious executables.



FORENSIC INVESTIGATION

- Finding the text file the attacker created
- File of Interest Located:
- File name: README.txt
 - Path: root/tools/README.txt
- Observation :
 - Timeline: 12-12-2015 (3:24:39)



REGISTRY FORENSICS

- Extracting system and user hives with FTK imager to perform Registry Forensics
 - SAM
 - SOFTWARE
 - SECURITY
 - SYSTEM
 - DEFAULT
 - NTUSER.DAT
 - USERCLASS.DAT



REGISTRY FORENSICS

- SAM is loaded in Registry Explorer to find the number of user accounts and login information
 - Path: SAM\Domains\Account\Users

Registry Explorer v2.0.0.0

FileToolsOptionsBookmarks (2/0)ViewHelp

Registry hives (1)Available bookmarks (2/0)

ValuesUser accounts

Enter text to search...Find

F:\RU\Fall 2024\Cyb...

CsTool-CreateHive-...

SAM

Domains

Account

Aliases

Groups

Users

000001F4

000001F5

000003E9

Names

Builtin

LastSkuUpgrade

RXACT

Unassociated deleted ...

Valid U...

User Id

Invalid ...

Total ...

Created On

Last Login Time

Last Password Change

Las...

Expi...

User Name

Full...

Pa...

Groups

Co...

Us...

Ho...

Int...

Re...

Ac...

Ho...

Pa...

Te...

No...

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W...

Se...

Pa...

Auto Locked

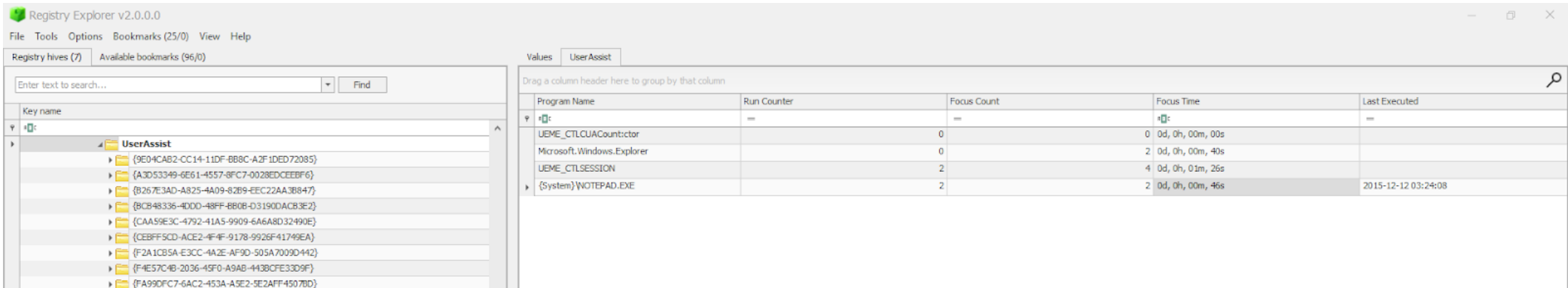
</

Observation: The Admin Account was logged in multiple times, which makes the user account suspicious



NTUSER.DAT

- ◆ Path:
NTUSER.DAT\Software\Microsoft\Windows\Currentversion\Explorer\UserAssist\{GUID}\Count



Registry Explorer v2.0.0.0

File Tools Options Bookmarks (25/0) View Help

Registry hives (7) Available bookmarks (96/0)

Enter text to search... Find

Key name

[-] HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist

[-] {9E04CAB2-CC14-11Df-BB8C-A2F1DED72085}

[-] {A3D53349-6E61-4557-8FC7-0028EDCEBF6}

[-] {B267E3AD-A825-4A09-82B9-EEC22AA3B847}

[-] {BCB48336-4DDD-48FF-8B0B-D3190DACB3E2}

[-] {CAA59E3C-4792-41A5-9909-6A6A8D32490E}

[-] {CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}

[-] {F2A1CB5A-E3CC-4A2E-AF9D-505A7009D442}

[-] {F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}

[-] {FA99DFC7-6AC2-453A-A3E2-5E2AFF4507BD}

Values UserAssist

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
[-] C:\Windows\WinSxS\x-wwp	==	==	[-] C:\Windows\WinSxS\x-wwp	==
UEME_CTLCUACountictor	0	0	0d, 0h, 00m, 00s	
Microsoft.Windows.Explorer	0	2	0d, 0h, 00m, 40s	
UEME_CTLSESSION	2	4	0d, 0h, 01m, 26s	
(System) \NOTEPAD.EXE	2	2	0d, 0h, 00m, 46s	2015-12-12 03:24:08

- ◆ Observation: The NTUSER.DAT file of the Administrator account, Notepad was being accessed at the time of the attack by the Administrator at 12-12-2015 3:24:08



SYSTEM ANALYSIS

System Information:

- Computer Name: SENSEI
- OS: Windows 8.1 Enterprise Edition,
- Time Zone: PST (Pakistan Standard Time)

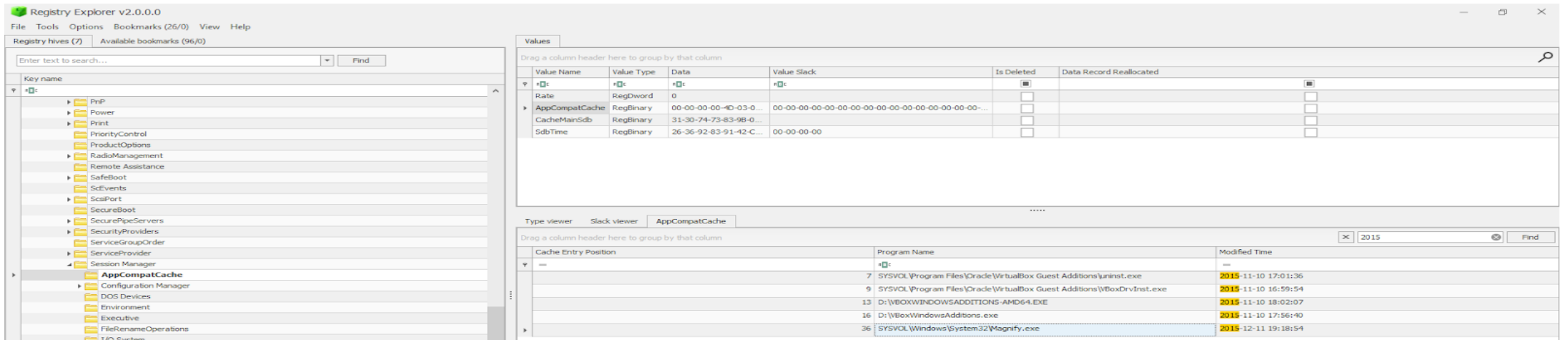
User Account Activities:

- Active User: Administrator
- Notepad.exe accessed during the attack.



SYSTEM

- **ShimCache /AppCompatCache:** Used to store the file name, file size, and last modified time of the executables.
- **Path :** SYSTEM\CurrentControlSet\Control\Session Manager\AppCompatCache



- **Observation: Magnify.exe was the only executable used close to the time of the attack**

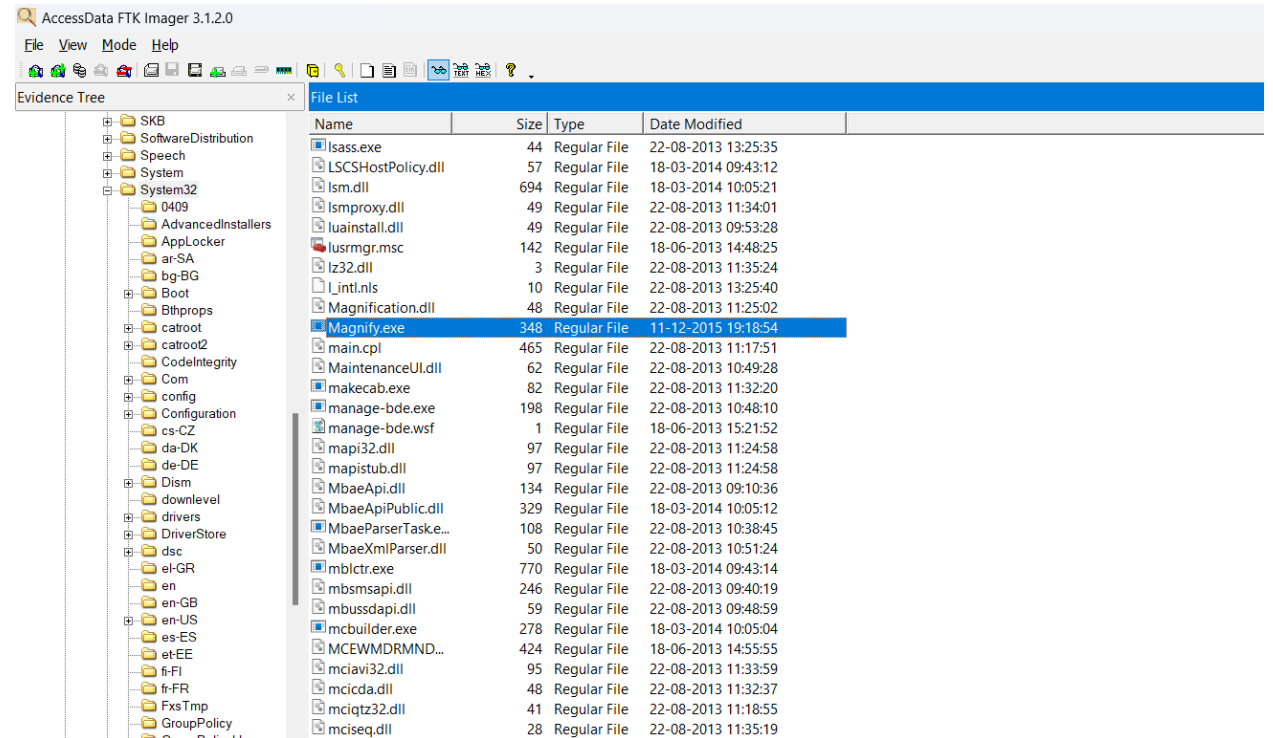


ANALYSIS

- Extracting Magnify.exe with the help of FTK imager to sandbox environment
 - Path: SYSVOL/Windows/System32/Magnify.exe

- Observations:

- Size: 346 KB
- Date Modified: 11-12-2015
- Time Modified: 7:18:54

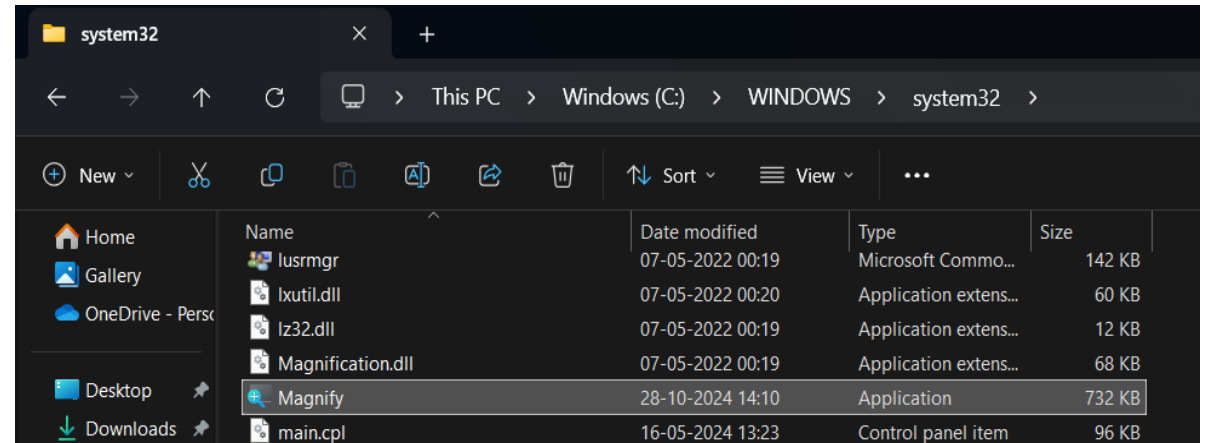
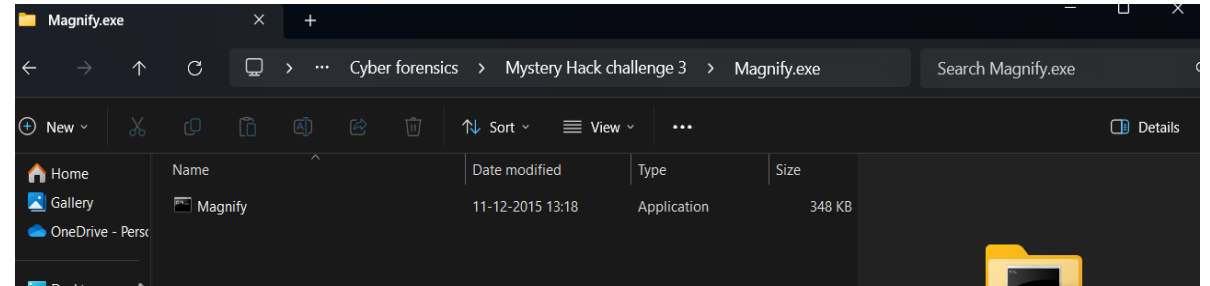


Name	Size	Type	Date Modified
lsass.exe	44	Regular File	22-08-2013 13:25:35
LSCSHostPolicy.dll	57	Regular File	18-03-2014 09:43:12
lsmdll	694	Regular File	18-03-2014 10:05:21
lsmpoxy.dll	49	Regular File	22-08-2013 11:34:01
luainstall.dll	49	Regular File	22-08-2013 09:53:28
lusrmgr.msc	142	Regular File	18-06-2013 14:48:25
tz32.dll	3	Regular File	22-08-2013 11:35:24
l_intl.nls	10	Regular File	22-08-2013 13:25:40
Magnification.dll	48	Regular File	22-08-2013 11:25:02
Magnify.exe	348	Regular File	11-12-2015 19:18:54
main.cpl	465	Regular File	22-08-2013 11:17:51
MaintenanceUI.dll	62	Regular File	22-08-2013 10:49:28
makecab.exe	82	Regular File	22-08-2013 11:32:20
manage-bde.exe	198	Regular File	22-08-2013 10:48:10
manage-bde.wsf	1	Regular File	18-06-2013 15:21:52
map32.dll	97	Regular File	22-08-2013 11:24:58
mapistub.dll	97	Regular File	22-08-2013 11:24:58
MbaeApi.dll	134	Regular File	22-08-2013 09:10:36
MbaeApiPublic.dll	329	Regular File	18-03-2014 10:05:12
MbaeParserTask...	108	Regular File	22-08-2013 10:38:45
MbaeXmlParser.dll	50	Regular File	22-08-2013 10:51:24
mbictr.exe	770	Regular File	18-03-2014 09:43:14
mbmsapi.dll	246	Regular File	22-08-2013 09:40:19
mbussapi.dll	59	Regular File	22-08-2013 09:48:59
mcbuilder.exe	278	Regular File	18-03-2014 10:05:04
MCEWMDRMND...	424	Regular File	18-06-2013 14:55:55
mcia32.dll	95	Regular File	22-08-2013 11:33:59
mcicda.dll	48	Regular File	22-08-2013 11:32:37
mcicq32.dll	41	Regular File	22-08-2013 11:18:55
mciseq.dll	28	Regular File	22-08-2013 11:35:19



INITIAL EVIDENCE

- Comparing the downloaded executable with the preexisting magnify.exe in system32 directory
- Observation
 - Difference in Size
 - Difference in Icon



CONFIRMATORY EVIDENCE

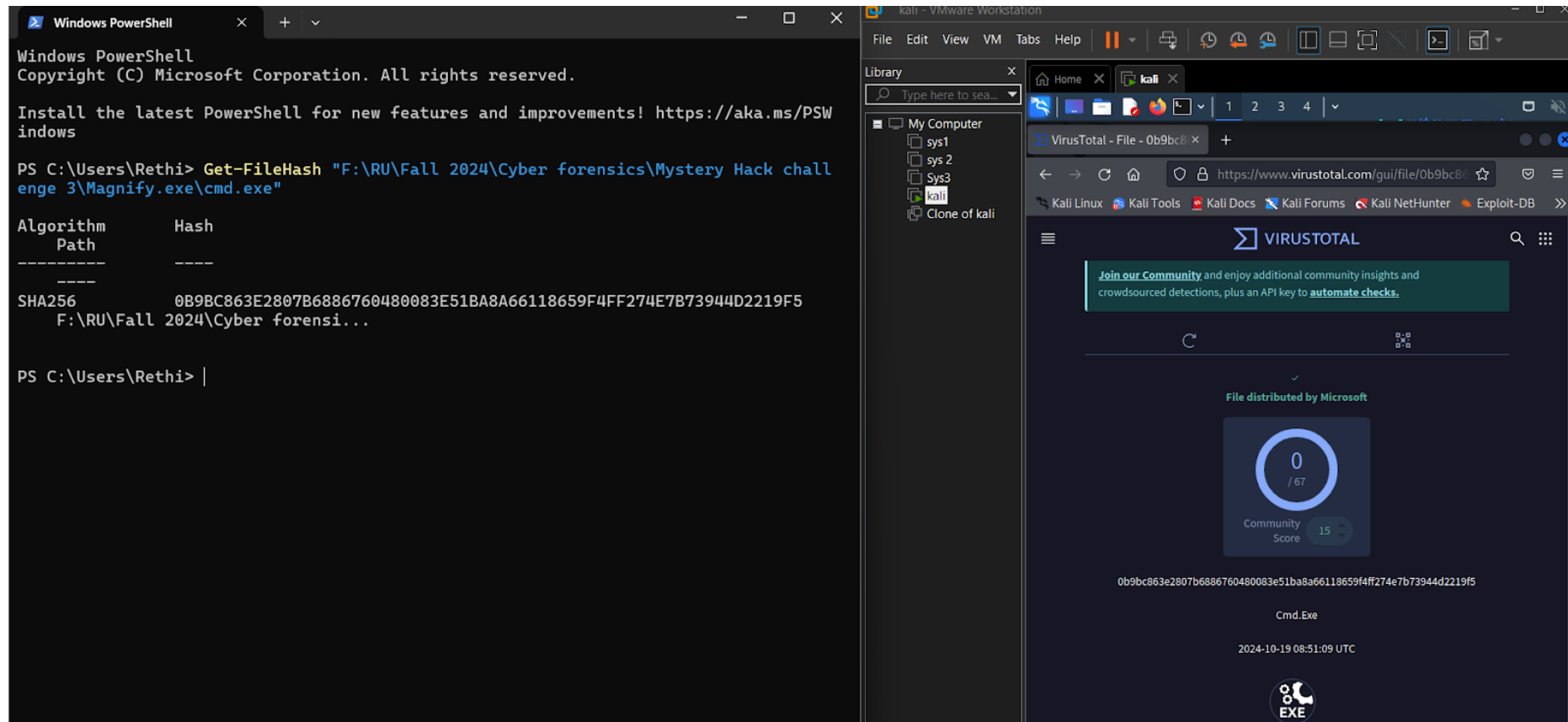
- Uploading Magnify.exe in VirusTotal
- Observation:
 - The file name is cmd
 - Obtained the hash of the file

 cmd	22-08-2013 05:03	Application	348 KB
 Magnify	11-12-2015 13:18	Application	348 KB

- Surprisingly Command Prompt And Magnify had the same icon and size



VERIFICATION



The image shows a Windows PowerShell terminal window on the left and a VirusTotal web interface on the right, both within a VMware Workstation environment.

Windows PowerShell Terminal:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSW
indows

PS C:\Users\Rethi> Get-FileHash "F:\RU\Fall 2024\Cyber forensics\Mystery Hack challenge 3\Magnify.exe\cmd.exe"

Algorithm      Hash
-----
SHA256         0B9BC863E2807B6886760480083E51BA8A66118659F4FF274E7B73944D2219F5
F:\RU\Fall 2024\Cyber forensi...
```

VirusTotal Web Interface:

The VirusTotal interface shows the file `0b9bc863e2807b6886760480083e51ba8a66118659f4ff274e7b73944d2219f5` (Cmd.Exe) distributed by Microsoft. The Community Score is 15/67. The file is identified as `Cmd.Exe` and was scanned on 2024-10-19 08:51:09 UTC.

SHA256

0B9BC863E2807B6886760480083E51BA8A66118659F4FF274E7B73944D2219F5



CONCLUSION

- Hypothesis: The attacker gained privileged access, and replaced Magnify.exe with cmd.exe.

Recommendations

- ◇ Enforce stricter access policies and implement multi-factor authentication.
- ◇ Regularly monitor and verify core executables' integrity.
- ◇ Strengthen firewall and intrusion detection systems.



QUESTIONS | FEEDBACK | FEELINGS

Thank you, guys

